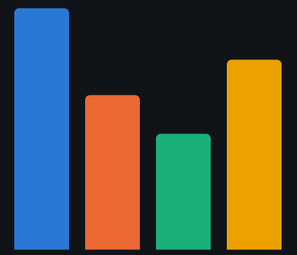
ADAPTIVE QEM BENCHMARK · PROJECT SUMMARY

When Does Quantum Error Mitigation Pay Off Under a Fixed Shot Budget?

Accounting, decision reliability, and operational boundaries of readout correction and zero-noise extrapolation, studied in noisy classical simulation.



Zhen-Jun Liu

November 2026 · github.com/Raina-ya/adaptive-qem-benchmark

All results simulated. No hardware result is claimed.

THE QUESTION

A better estimate is not the goal. A better decision is.

Error mitigation removes bias but spends extra circuits and inflates variance. Most benchmarks score the mitigated *value*. In practice the value feeds a *choice*: which candidate, which parameter, which circuit is best.

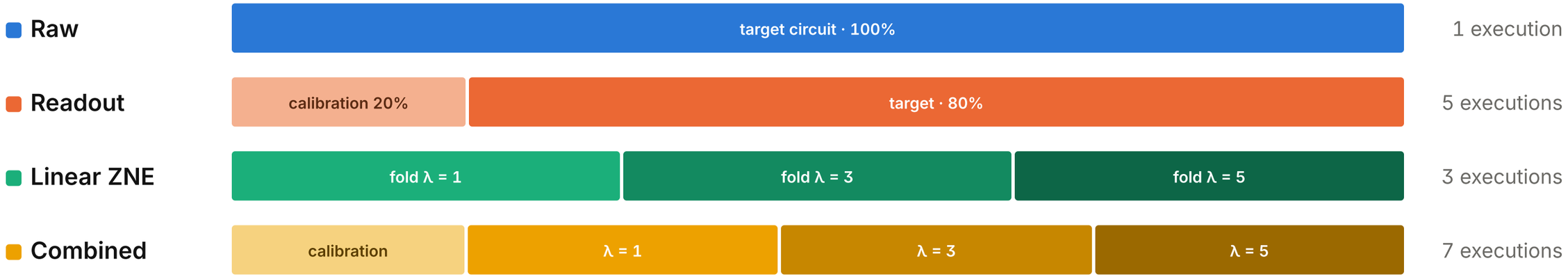
- **Q1.** When does mitigation improve estimation at a fixed total budget?
- **Q2.** When does better estimation produce a better decision?
- **Q3.** Can runtime evidence tell when to trust mitigation, an extrapolation, or a certificate?
- **Q4.** Where do readout correction, calibration reuse, and adaptive selection stop working?

> Thesis: under a fixed budget, QEM improves expectation RMSE but does not guarantee decision reliability. Whether it pays off depends on error structure, signal gap, calibration cost, variance amplification, and model mismatch.

METHOD · FIXED-BUDGET ACCOUNTING

Every method spends the same shots, and every overhead is charged

Calibration, folded executions, pilots, and diagnostic circuits all come out of one declared budget. Failed corrections stay in the data as wrong decisions.



Execution counts at width 2 and 16,384 shots. Raw, readout, and ZNE splits are exact; the combined split is schematic.

Linear ZNE intercept weights (13, 4, −5)/12 ⇒ variance × 4.375, standard deviation × 2.09.

DESIGN

Ten studies, each frozen in the repository before its run

STUDY	SCALE	RULES
Main grid	144 cells · 50 repeats	With results
Combined QEM	6 cells · 2 budgets	Committed first
QAOA selection	3 graphs · 5 candidates	Committed first
Runtime selector test	160 fresh cells	Committed first
Signal strata	126 circuits · 7 noises	Committed first
ZNE validity gate	256 candidates · holdout	Committed first
Confirmatory	34 + 72 comparisons	Committed first
Shots to reliability	40 cells · 4 methods	Committed first
Selector v3 · certification	72 tasks · 4 families	Committed first
Readout boundaries	widths 4–10	Committed first

- Rules chosen on training and validation splits; each holdout read once.
- Raw repeat-level rows are never overwritten.
- Seeds derive from one root seed.
- “Committed first” means repository order. It is not an independently timestamped preregistration.

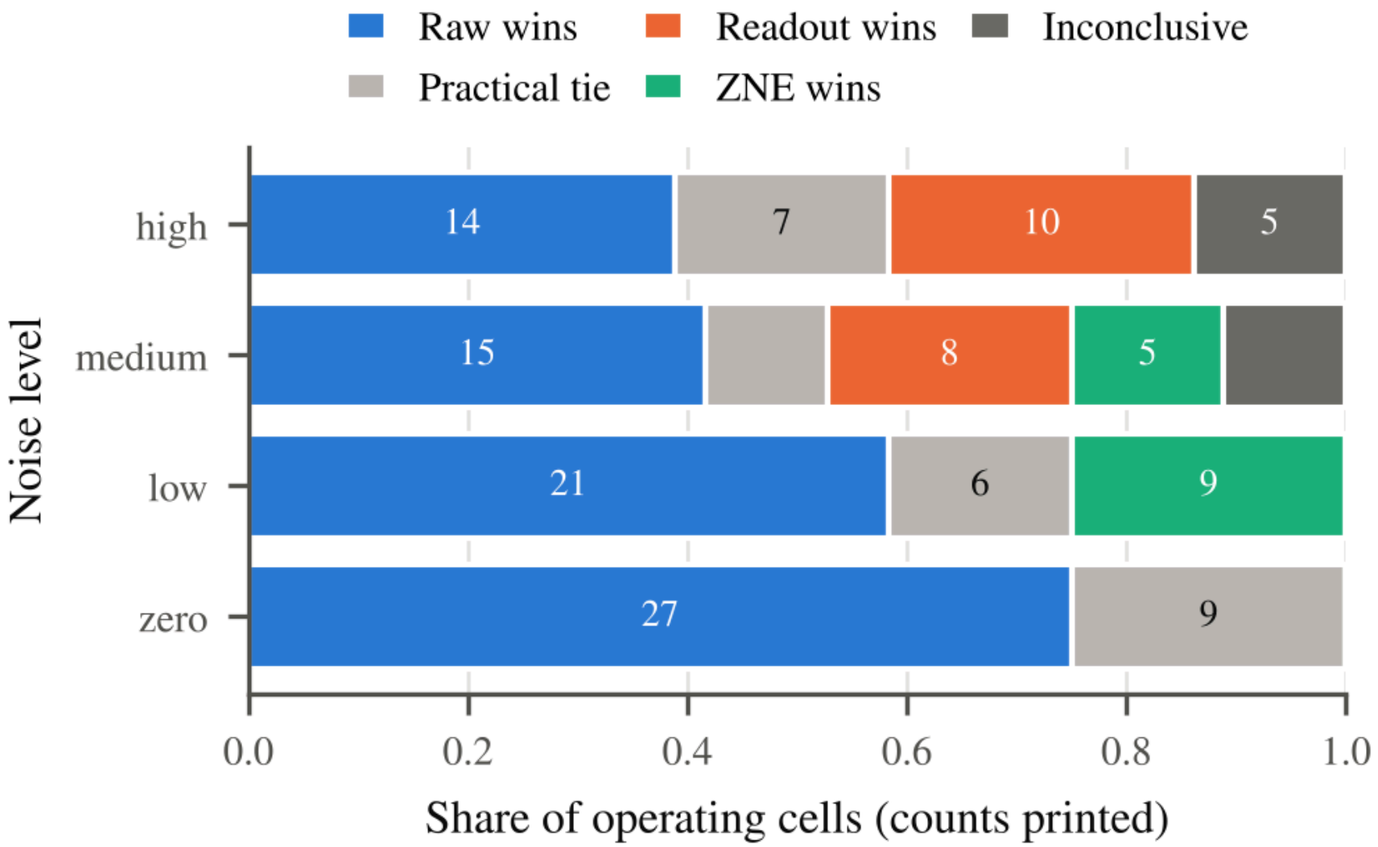
01

Fixed-budget benchmarking

Once every overhead is charged, no method dominates.

FINDING 1 · OPERATING REGIMES

No method wins everywhere, and raw mostly wins by small margins



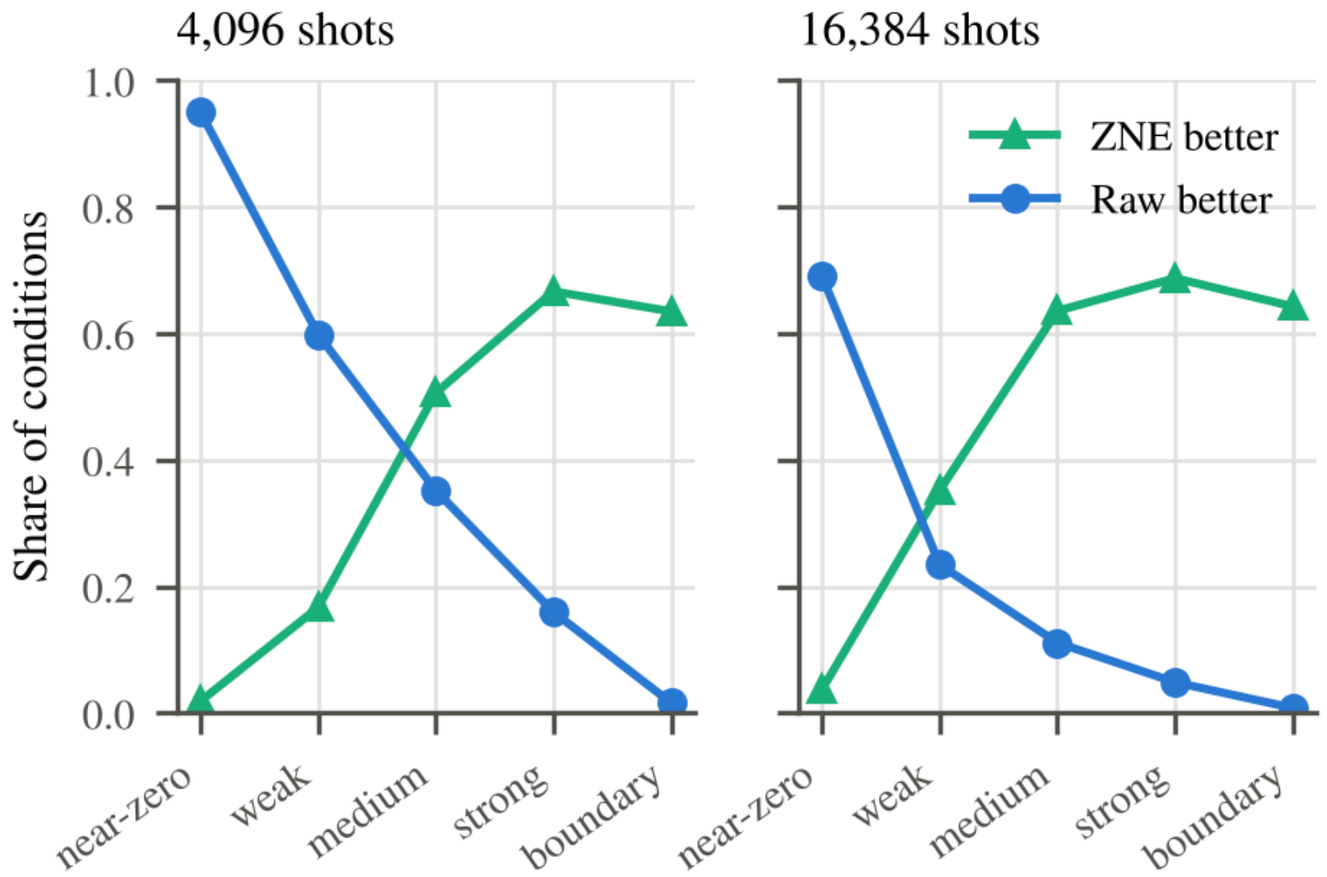
THRESHOLD	RAW	TIE	READOUT	ZNE
0.002	83	16	19	17
0.005 (frozen)	**77**	**26**	**18**	**14**
0.010	46	62	18	10
0.020	12	100	18	6

Cells of the 144-cell grid by label as the practical-equivalence threshold varies; 8–9 cells are inconclusive at every threshold. Sensitivity analysis added after the results were known.

Readout wins are stable (18–19). Raw “wins” shrink from 77 to 12 as the threshold grows.

FINDING 1 · SIGNAL STRENGTH

ZNE gains track the attenuation it can remove, not signal size



95.1%

of near-zero conditions where raw is better (4,096 shots). Its median bias there is 43% of the reference.

63.5%

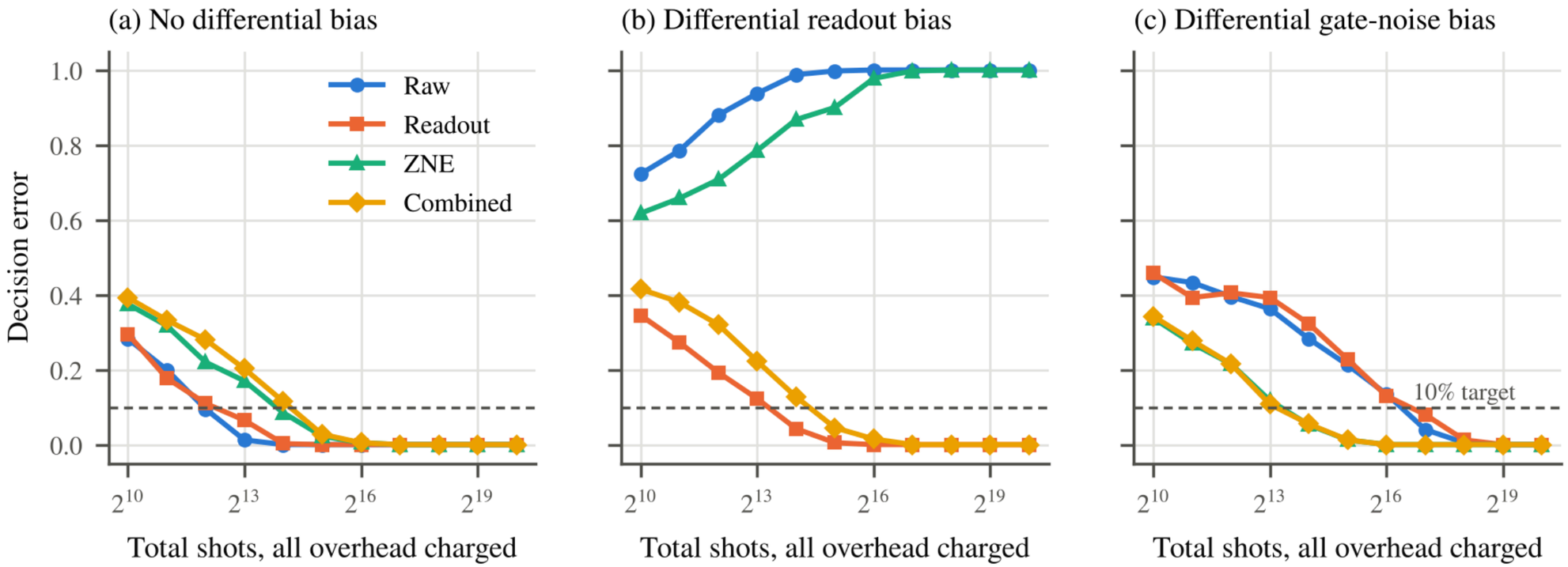
of boundary-adjacent conditions where linear ZNE is better.

0.611

difference in ZNE win share, boundary minus near zero; circuit-cluster interval 0.545 to 0.674, on 126 circuits frozen by noiseless value.

FINDING 1 · SHOTS TO RELIABILITY

Shots cannot buy off a differential bias



AT A 10% DECISION-ERROR TARGET, 40 CELLS	RAW	READOUT	ZNE	COMBINED
Reached by some budget up to 2^{20}	26	33	27	34
Blocked by bias (error rises with shots)	13	7	11	5

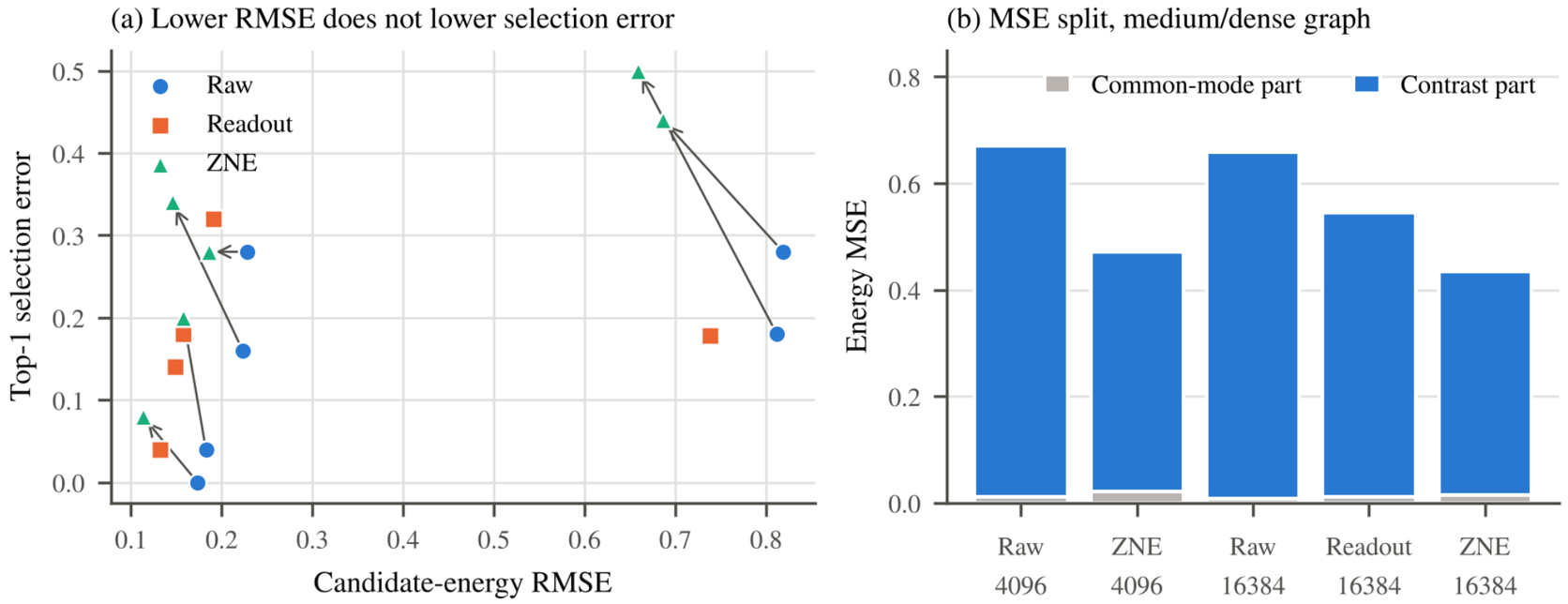
02

Estimator quality versus decision reliability

A method can lower RMSE and still choose worse.

FINDING 2 · QAOA CANDIDATE SELECTION

ZNE lowers energy RMSE yet picks the wrong candidate more often



0.271 vs **0.379**

selection error, raw vs ZNE, on new repeats medium/dense graph, 16,384 shots

398 pairs · p = 0.001

confirmatory test, planned with stated power and Holm correction

0.153

correlation between RMSE and selection error across repeats

FINDING 2 · MECHANISM

The decision depends on the decisive gap, and ZNE inflates its spread

- Energy MSE splits into a common-mode term and a contrast term. Selection depends only on contrasts.
- On the medium/dense graph, **77%** of the MSE that ZNE removes comes from one control candidate that never contends for the top.
- The top-two gap: ZNE shrinks its mean from **0.045 to 0.016** and widens its spread from **0.053 to 0.137**.
- The normal approximation predicts inversion rates of **0.20 and 0.45**; observed **0.18 and 0.50**.

REJECTED EXPLANATION

“Raw rankings are stable because raw errors are shared.”

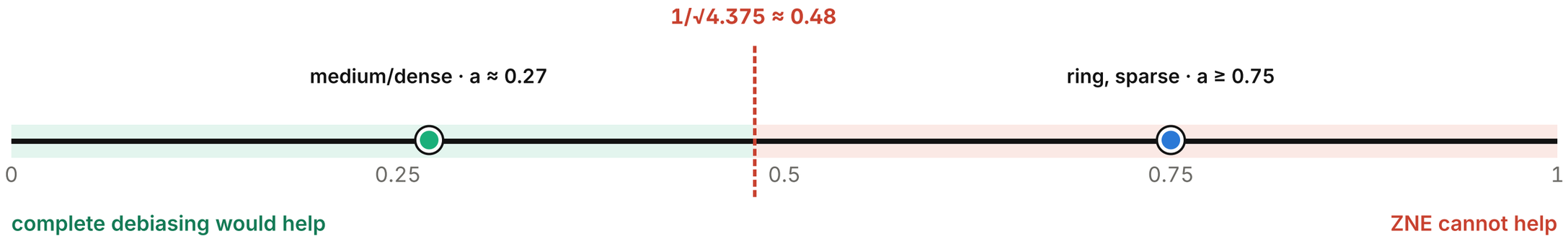
Off-diagonal error correlations across repeats are near zero for both raw and ZNE. The inversion comes from the gap’s variance, not from shared raw error.

THEORY · PROPOSITION 1

Debiasing helps a decision only if it outgrows its own noise

$$\Delta_m > \sqrt{\gamma}\,\Delta_r$$

Δ_r, Δ_m: mean decisive gap of raw and mitigated estimates · γ: variance amplification at equal total budget · normal approximation, $\Delta_r > 0$



Raw attenuation a = Δ_r/Δ. The preference does not depend on the budget. The medium/dense graph could have benefited but ZNE achieved a gap ratio of 0.36. The threshold was derived after these data, so it is a hypothesis for new graphs.

03

Validity and operational boundaries

The runtime checks we tested see sampling error, not differential bias.

FINDING 3 · RUNTIME TOOLS

Gates, selectors, and certificates control noise, not bias

ZNE VALIDITY GATE V2

False rejection 0.000, false acceptance 0.176

Chosen from 256 runtime-only candidates and scored once on 30 held-out circuits. Its slope check measures whether decay is resolved, not whether the extrapolate is right, so false acceptance rises with shots.

SELECTOR V3 · NO ROUTE PASSED

Regret 0.0031, the same as always-readout

At 1.63× the two-qubit gate executions and 0.64 coverage. Always-raw had the lowest regret (0.0024): candidates share one readout, so its bias cancels in the comparison.

CERTIFICATION FAILS ON NEAR TIES

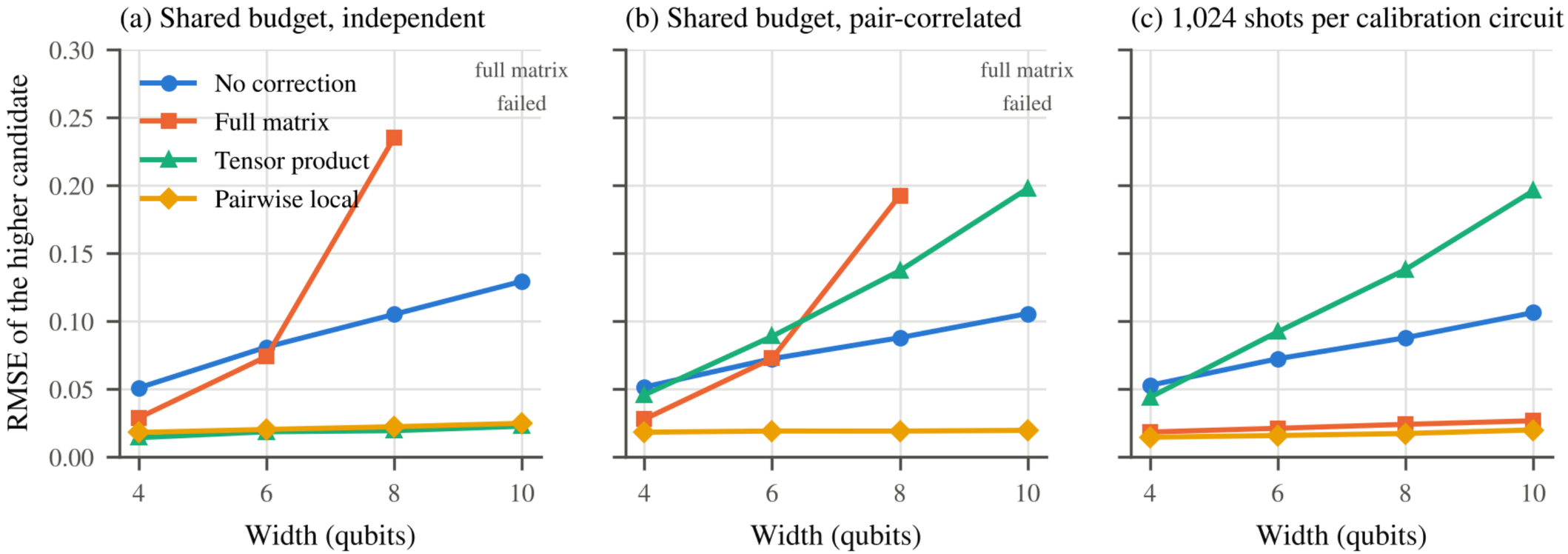
35.9% of certified near-tie runs are wrong

Corrected looks keep false certification at 1.7–2.7% at zero margin, against 22.4–24.9% for peeking. But a biased raw estimator certifies the wrong winner in 100% of repeats after about 2,000 shots.

> Every interval-based tool controls the σ term in $P_{err} \approx \Phi(-(\Delta+\beta)\sqrt{N}/\sigma)$. None of them sees β.

FINDING 3 · READOUT AT SCALE

A mismodeled readout correction does worse than none



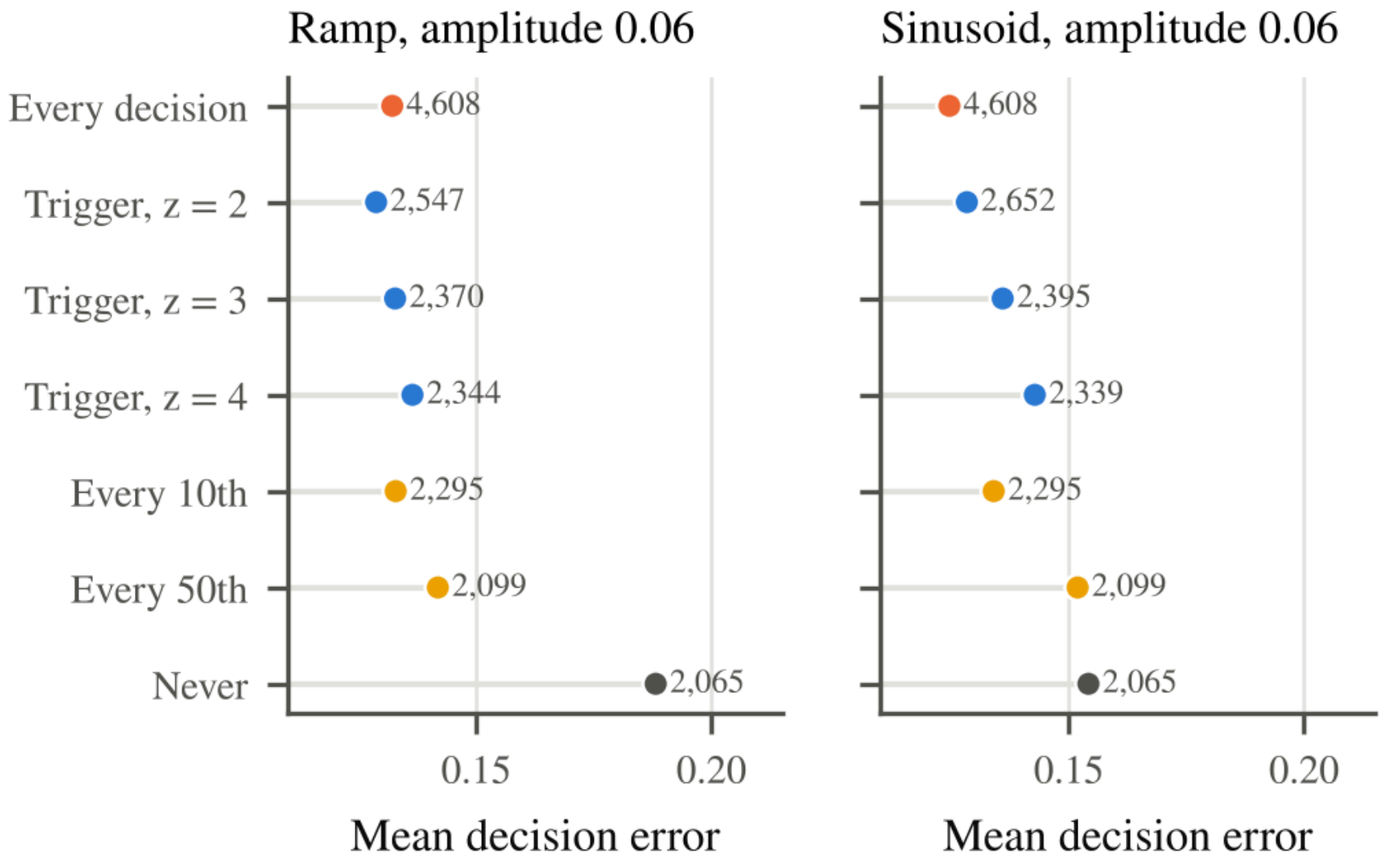
Resource failure. Full-matrix correction fails in 93–100% of width-8 runs under a shared 4,096-shot budget; with enough calibration it completes, using 80–98.5% of all shots.

Model failure. Under pair-correlated readout, tensor-product, subspace, and matrix-free corrections become worse than raw. Pairwise-local correction stays below 0.05.

Blind warning. A 1,024-shot check warns in time in 76.2% of mismatch cases, but not when flips become correlated with equal marginals. Condition number never warns.

FINDING 3 · CALIBRATION REUSE AND HARDWARE

Triggered recalibration works; hardware is a documented no-go



≈ 60% of the shots

A check-triggered policy at z = 2 matches every-decision recalibration in mean decision error, except under sinusoidal drift at amplitude 0.06.

1 of 7

go requirements met in the hardware readiness review; all 6 no-go triggers apply. The plan would need about 72.1 million shots. No job was submitted.

NEGATIVE RESULTS, CLASSIFIED

Every failure is kept and named by mechanism

RESULT	TYPE	MORE SHOTS FIX IT?
Lower RMSE, worse QAOA selection	METHOD	No: variance × 4.375
Biased method misses a decision target	METHOD	No: error rises
ZNE gate false acceptance	METHOD	No: rises with shots
Near-tie false certification	METHOD	No
Adaptive selectors v1, v2, v3	RESOURCE	Unknown
Full matrix fails at width 8	RESOURCE	Yes, at 80–98.5% calibration share
Independent model, correlated readout	MODEL	No
No steep decision-error rise found	DESIGN	Inconclusive

Method failures persist with unlimited shots. **Resource** failures disappear with more calibration or target shots. **Model** failures come from a wrong noise or readout model.

> Mitigation pays off when it removes a bias that differs between the alternatives and outgrows its own variance cost.

VERIFICATION

Every printed number traces to code and is recomputed independently

195

printed numbers checked by the claim ledger, 0 mismatches

23

computed claims recomputed without importing project code

12 of 16

long generators reproduced in a clean Linux rebuild

32

references, each checked for authors, venue, and DOI

- Artifact registry, content hashes, run logs, and one-job-per-generator clean rebuilds.
- Four generators differ between Windows and Linux. Three change no result (a signed zero, an infinite condition number, an unused rate). In 29 of 5,760 scalable-readout runs, convergence depends on the platform. Recorded as findings; no tolerance was loosened.
- Nine tables and the figures are source-traced but not all recomputed, so the paper as a whole is **not independently verified**.

LIMITS AND NEXT STEPS

What this does not show, and what comes next

LIMITS

Existence, not a law

Synthetic noise only. Three curated QAOA graphs, 30 holdout circuits, 20 selector tasks. Only four method families. Sequence of studies chosen adaptively; only the confirmatory set is confirmatory at project level.

NEXT

Detect differential bias

Extrapolate the pairwise gap directly, pair each candidate with a Clifford twin of known value, and test randomized compiling against coherent over-rotation.

THEN

Predict before measuring

Register Proposition 1 predictions on 50+ new graphs before running them, add device-informed noise models, and use the open-plan budget for one readout-correlation experiment.

> Judge a mitigation method by the decision it serves, not only by the error of the value it returns.